

Indices, Surds & Quadratics



Laws of indices

- 1 Simplify $x^5 \times x^{-2}$.
 - 2 Simplify $\left(\frac{8x^6}{27y^3}\right)^{1/3}$.
 - 3 Simplify $\frac{x^{-3} \times x^7}{x^2}$, giving your answer as a single power of x .
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Solving exponential equations

- 4 Solve $3^{2x-1} = 81$.
 - 5 Solve $4^x = 8^{x-1}$, writing both sides with base 2 first.
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Simplifying surds

- 6 Simplify $\sqrt{75}$.
 - 7 Simplify $\sqrt{20} + \sqrt{45}$.
 - 8 Simplify $\sqrt{48} - \sqrt{12} + \sqrt{3}$.
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Rationalizing the denominator

- 9 Rationalize $\frac{6}{\sqrt{3}}$.
 - 10 Rationalize $\frac{5}{2-\sqrt{3}}$ by multiplying by the conjugate.
 - 11 Rationalize $\frac{4}{\sqrt{7}+1}$.
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Completing the square

- 12 Write $x^2 - 8x + 10$ in the form $(x - a)^2 - b$, stating a and b .
- 13 Use completing the square to find the minimum value of $f(x) = 2x^2 - 12x + 7$.

- 14 Use completing the square to solve $x^2 + 6x - 7 = 0$.
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Solving quadratics: factoring and the formula

- 15 Solve $x^2 - 2x - 15 = 0$ by factoring.
- 16 Solve $3x^2 + 2x - 4 = 0$ using the quadratic formula, giving exact answers.
- 17 Solve $x^2 - 6x + 9 = 0$, and state what is special about the solution.
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The discriminant

- 18 Use the discriminant to determine the number of real roots of $x^2 + 4x + 7 = 0$.
- 19 Find the values of k for which $x^2 + kx + 9 = 0$ has exactly one (repeated) real root.
- 20 Show that $2x^2 + 3x + 5 = 0$ has no real roots, using the discriminant.
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Quadratic equations disguised in another variable

- 21 Solve $x^4 - 13x^2 + 36 = 0$ by substituting $u = x^2$.
- 22 Solve $2^{2x} - 5(2^x) + 4 = 0$ by substituting $u = 2^x$.
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Sum and product of the roots

- 23 For the quadratic $2x^2 - 7x + 3 = 0$, find the sum and product of its roots without solving for them, using $\alpha + \beta = -\frac{b}{a}$ and $\alpha\beta = \frac{c}{a}$.
- 24 A quadratic equation has roots $\alpha = 4$ and $\beta = -3$. Find a quadratic equation with these roots, in the form $x^2 - (\alpha + \beta)x + \alpha\beta = 0$.
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A well-designed word problem: optimizing area

- 25** This question has two parts. A farmer has 60 meters of fencing to enclose a rectangular field against an existing straight wall (so fencing is only needed on three sides). Let the width perpendicular to the wall be x meters. (a) Show that the enclosed area is $A = x(60 - 2x)$. (b) Use completing the square to find the value of x that maximizes the area, and state the maximum area.